

ECDH

Alice $E, \#E, G$ Bob

~~at~~

$$a \in \{1, 2, \dots, \#E-1\} \quad b \in \{1, 2, \dots, \#E-1\}$$

$$A = aG \xrightarrow{aG} A$$

$$\cancel{B} \quad B \xleftarrow{\quad} B = bG$$

$$k_{ab} = aB$$

$$k_{ab} = bA$$

Point compression

- for every x there are 2 y values
do not need to transport
full (x, y)
- Each point contains a unique
 x and one of two y
- $y = \sqrt{x^3 + 2x + \alpha} \pmod{p}$
- The app a bit to represent
a positive or -ve y